

Design and Implementation of a Pedestrian Traffic Light Controller Using Arduino Uno and Finite State Machines

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Abstract—This paper presents the design and implementation of a pedestrian traffic light control system using an Arduino Uno platform. The system addresses a real-world safety issue in Copiapó, where pedestrians frequently cross a non-authorized area, increasing accident risk. A finite state machine was developed to control vehicular and pedestrian signals efficiently. The system includes a pedestrian request button, a night mode, and timing optimization to minimize traffic disruption. A cost analysis and system performance evaluation are also presented.

Keywords—*Arduino, Finite State Machine, Traffic Control, Embedded Systems, Moore Machine*

I. INTRODUCTION

In urban areas such as Copiapó, pedestrian safety is a critical concern, especially in high-traffic avenues like Copayapu. Many pedestrians cross at unauthorized locations, exposing themselves to accidents. To mitigate this issue, a pedestrian traffic light system is required to regulate crossing while minimizing disruption to vehicular flow.

This work proposes the design and implementation of an automated traffic light controller using an Arduino Uno. The system is based on a finite state machine model, allowing structured control of signal transitions. Additionally, constraints such as limited traffic interruption, night operation, and pedestrian request handling are considered.

II. METHODOLOGY

A. Requirements

The system is designed to address unsafe pedestrian crossings in a high-traffic urban area.

The following requirements were established:

- Ensure pedestrian safety during road crossing
- Minimize disruption to vehicular flow
- Prevent excessive activation of pedestrian lights through a restriction mechanism
- Design and implement a night operation mode
- Provide a low-cost implementation

- Define appropriate timing for vehicular and pedestrian signals
- Implement the system using a microcontroller-based implementation

B. System Design

The system consists of:

- Vehicular traffic lights (red, yellow and green)
- Pedestrian traffic lights (red and green)
- Pedestrian request button
- Night mode activation button

The controller manages traffic flow based on predefined timing and user interaction. The night mode activation via a button represents a simplified approach, as this functionality could be automated using a time-based variable in a real-world implementation.

C. Software Development

The control system was prototyped on the Thinkercad platform using C/C++, simulating an Arduino Uno board and structured as a finite state machine. A switch-case structure was used to represent each state, defining both outputs and transition conditions.

State transitions depend on timing, pedestrian requests, and night mode activation. Timing control is handled using the `millis()` function.

To prevent excessive interruption of vehicular traffic, a restriction mechanism based on a cycle counter was implemented. The system only processes pedestrian requests after a predefined number of completed traffic cycles.

Night mode is activated through an input signal and modifies the system behavior to a blinking yellow light for vehicles while maintaining a red signal for pedestrians.

D. Validation and Testing

The system was tested in simulation using the Thinkercad platform, validation included:

- Correct state transitions
- Proper response to the pedestrian request button activation
- Proper response to the night mode button and the mode itself
- Accurate timing behavior

E. Hardware Implementation

Once the prototype behaved as intended, the system was implemented on the real board using:

- Arduino Uno
- 5 LEDs, representing the traffic lights
- Resistors
- 2 buttons
- Breadboard and wiring

The implemented system performed as expected

III. SYSTEM DESIGN AND MODELING

A. States

The finite state machine consists of five states:

- S0 (000): Green vehicular traffic light and red pedestrian traffic light on.
- S1 (001): Yellow vehicular traffic light and red pedestrian traffic light on.
- S2 (010): Red vehicular traffic light and green pedestrian traffic light on.
- S3 (011): All vehicular traffic lights off and red pedestrian traffic light on.
- S4 (100): Yellow vehicular traffic light and red pedestrian traffic light on.

Being S0, S1 and S2 states of the normal mode, and S3 and S4 states of the night mode.

B. Conditions/Inputs

The system's behavior is represented using a Moore diagram, where transitions depend on:

- M: Night mode
- P: Pedestrian request
- T: Time condition

These inputs define the transition conditions of the system within the Moore model. The variable M has priority, as it forces the system into night mode operation regardless of other inputs. The pedestrian request P allows users to trigger a crossing sequence, subject to the restriction mechanism implemented in

the system. Finally, T represents the timing condition associated with each state, enabling transitions once the predefined time intervals have elapsed.

C. State Table

Given the states previously defined, the conditions and the program developed, the following state table was devised:

TABLE 1. STATE TABLE

Input			Current State	Next State	Output				
M	P	T			VV	AV	RV	VP	RP
1	X	X	000	011	1	0	0	0	1
0	0	0	000	000	1	0	0	0	1
0	0	1	000	001	1	0	0	0	1
X	1	X	000	001	1	0	0	0	1
1	X	X	001	011	0	1	0	0	1
0	X	1	001	010	0	1	0	0	1
0	X	0	001	001	0	1	0	0	1
1	X	X	010	011	0	0	1	1	0
0	X	1	010	000	0	0	1	1	0
0	X	0	010	010	0	0	1	1	0
0	X	X	011	000	0	0	0	0	1
1	X	1	011	100	0	0	0	0	1
1	X	0	011	011	0	0	0	0	1
0	X	X	100	000	0	1	0	0	1
1	X	1	100	011	0	1	0	0	1
1	X	0	100	100	0	1	0	0	1

Where the outputs stand for:

- VV: Green vehicular traffic light
- AV: Yellow vehicular traffic light
- RV: Red vehicular traffic light
- VP: Green pedestrian traffic light
- RP: Red pedestrian traffic light

Acronyms of their respective description in Spanish.

Table 1 presents the complete state table of the system, defining the relationship between inputs, current states, next states, and outputs. The inputs correspond to night mode (M), pedestrian request (P), and timing condition (T), while the states are encoded in binary form. Each row specifies the transition rules of the finite state machine, including the use of "don't care" conditions (X) to simplify the representation. The outputs follow a Moore model, meaning they depend exclusively on the current state and are therefore constant within each state.

D. Finite State Machine

The diagram in Figure 1 shows all possible transitions between states. It follows the Moore machine model, where output depends solely on the current state. Each state is represented as a node labeled with its corresponding outputs, while transitions between states are defined by input conditions such as timing, pedestrian requests, and night mode activation.

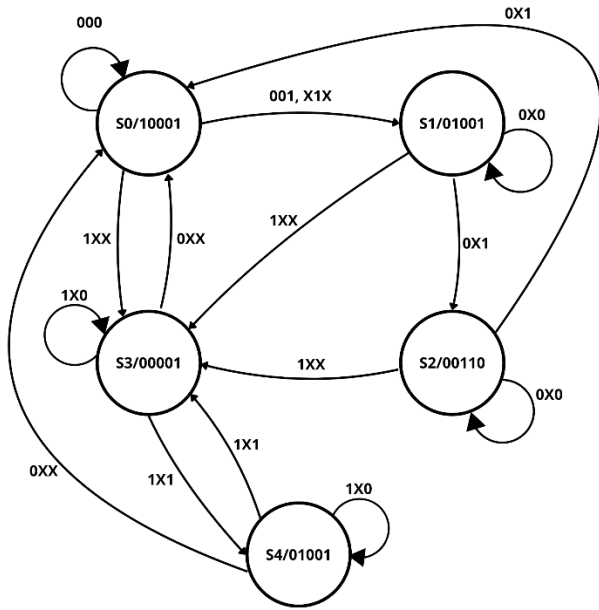


Fig. 1: Moore state machine diagram

E. Control Logic

The system operates in two main modes: normal mode and night mode. In normal mode, the traffic light follows a cyclic sequence of states defined as $S0 \rightarrow S1 \rightarrow S2 \rightarrow S0$. During this operation, vehicular traffic is prioritized, with the system maintaining its cycle continuously. The pedestrian crossing phase (S2) is part of the normal cycle; however, its activation can also be influenced by a pedestrian request, provided certain conditions are met. A restriction mechanism based on a cycle counter ensures that at least three full cycles occur before a new request is accepted, preventing excessive interruptions to vehicular flow.

For demonstration purposes, the timing of each state was configured with reduced intervals, specifically 5 seconds for the green vehicular light (S0), 3 seconds for the yellow light (S1), and 5 seconds for the pedestrian crossing phase (S2). These values allowed rapid observation of the system behavior during testing. In a real-world scenario, longer durations would be required; for instance, a vehicular green time of 30-40 seconds and a pedestrian crossing interval of approximately 10-15 seconds would provide safer and more realistic operation, depending on road conditions and pedestrian flow and considering Copayapu is the main road of Copiapó.

In night mode, the system changes to a simplified operation in which it alternates between states S3 and S4. This produces a blinking yellow light for vehicles, while the pedestrian signal remains red.

IV. IMPLEMENTATION

The system was physically implemented using the previously described hardware components. The Arduino board controls the LEDs according to the finite state machine logic.

Table 2 presents the mapping between Arduino pins and the corresponding traffic light outputs and button inputs.

TABLE 2. PIN CONFIGURATION

Arduino Pin	Description
5	VV: Green vehicular traffic light
6	AV: Yellow vehicular traffic light
7	RV: Red vehicular traffic light
11	VP: Green pedestrian traffic light
13	RP: Red pedestrian traffic light
8	Pedestrian Button
9	Night Mode Button

V. COST ANALYSIS

Table 3 details the cost of the solution:

TABLE 3. COST ANALYSIS

Item	Quantity	Cost (CLP)
Arduino Uno	1	\$24990
LEDs, resistors, push buttons, breadboard	1 set	\$7990
DuPont connectors	1 set	\$2990
System design (16 hours of work at \$15000 per hour)	1	\$240000
Installation	1	\$40000
Total	-	\$315970

The total estimated cost of the solution is \$315970 Chilean pesos. Most of the cost corresponds to the system design, while the hardware components, even when bought as a set, represent a small part of the total. This is because labor and expertise are the main factors in the development of this kind of solution.

Considering the small scale of the implementation of the solution, it still can be considered cost effective.

VI. RESULTS

The system was successfully prototyped, tested, and later implemented on a physical Arduino board. The observed behavior matched the expected operation defined by the finite state machine, confirming the correctness of the design and implementation.

The timing configuration ensured adequate intervals for pedestrian crossing while minimizing unnecessary interruptions to vehicular traffic. The restriction mechanism operated as intended, allowing controlled activation of pedestrian requests.

Overall, the system demonstrated stable and reliable performance.

VII. CONCLUSION

This work presented the design and implementation of a traffic light control system based on a finite state machine using an Arduino platform. The proposed solution addresses the problem of unsafe pedestrian crossings while maintaining efficient vehicular traffic flow.

The use of a Moore machine model enabled a structured design, ensuring alignment between the theoretical model, the state table, and the implemented system. Additionally, the inclusion of a restriction mechanism improved system performance by limiting unnecessary pedestrian interruptions.

REFERENCES

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